

Pratyusha Adibhatla

✉ pratyushaadibhatla3@gmail.com ☎ +1 (682) 374-3740 in LinkedIn GitHub Portfolio
H4 EAD(NO SPONSORSHIP REQUIRED) — Current Location: Santa Clara, CA — Relocation: Yes

Summary

Data Engineer with 3+ years building production ETL/ELT pipelines across AWS and Azure, processing 1M+ monthly financial transactions with PySpark, advanced SQL, and Databricks/Spark. Strong in big-data distributed processing, automated data-quality and schema-drift validation, dimensional data modeling (Star/Snowflake), and Git/CI-driven pipeline delivery in production.

Experience

Data Engineer, Finta

May 2026 – Present

- Built end-to-end student verification in Next.js/React .edu email code flow, document-upload fallback, status tracking, and approval/rejection emails — with server-side validation and hardened Supabase RLS so users can't self-approve.
- Shipped a responsive admin review dashboard and gated Stripe student pricing: approve/reject flows replacing manual SQL review, server-side approval checks before checkout, and a verified-only billing CTA Tailwind-styled for desktop/mobile, with post-deploy monitoring of approval and checkout flows in production.
- Own Aurora's (LLM agent) document-intelligence and retrieval layer end to end: a hybrid semantic + keyword search-and-ranking system fused with reciprocal rank fusion in Postgres/pgvector, plus a multi-format data export pipeline (PDF, DOCX, CSV, Markdown).
- Designed data models and Python backend services for 0-to-1 features; enforce data integrity via server-side validation and hardened Postgres row-level security, delivered through Git PR review with automated CI/CD.

Data Engineer, UCSD Wireless Communication, Sensing, and Networking Lab

Jan 2025 – Jul 2025

- Orchestrated ETL pipelines with Azure Data Factory orchestrating PySpark jobs on Azure Databricks, cutting LiDAR storage 1.2TB (85%) via automated 3D point-cloud downsampling and Parquet output to ADLS.
- Cut analytics-readiness time 80% and query times 75% by architecting an end-to-end Medallion Architecture (Bronze/Silver/Gold) on ADLS, turning multi-step manual extraction into one-click, analysis-ready DataFrames.
- Eliminated 15+ hours of manual data wrangling per week across the research team by deploying a dynamic categorization engine that parses schema metadata to achieve 100% automated segregation of datasets by sensor type (LiDAR, Radar, WiFi).

Research Assistant - Data Engineer – Retrieval Systems, University at Buffalo

Sep 2024 – Dec 2024

- Established a multimodal vector search pipeline to ingest and structure unstructured data (video, image, text) — applying computer vision (OCR) and transcription to convert images and video into retrievable text for low-latency RAG.
- Optimized data indexing workflows in ChromaDB by designing metadata schema, regenerating embeddings, and refreshing vector indexes when new data was added to ensure consistency across modalities.

Associate Data Engineer, Cognizant (Client: Liberty Mutual Insurance)

Dec 2020 - Jun 2022

- Optimized AWS EC2 ETL pipelines processing 1M+ monthly financial (insurance) transactions; monitored pipeline health in Datadog and held SLA adherence by resolving 20+ weekly job failures via log analysis, S3 dependency checks, and DB lock monitoring.
- Orchestrated advanced SQL transformations across Amazon RDS and Redshift, reducing financial reconciliation runtime and resolving cross-system inconsistencies and stranded payment records using DynamoDB metadata.
- Built automated data-quality validation frameworks detecting schema drift, null anomalies, and reconciliation inconsistencies — reducing data errors 26% across audit-compliant financial transactions.
- Collaborated cross-functionally with client stakeholders to translate data requirements into technical solutions, managing custom data requests and production incidents via **Jira** workflows, reducing manual triage time by 40%.

Projects

Job Recommender System – LinkedIn style & Analytics

Sep 2023 – Dec 2023

- Built batch ETL pipelines in Apache Spark using DataFrame APIs, window functions, and Parquet to transform job posting and user interaction data into analytics-ready datasets.
- Executed a batch recommendation workflow in Spark combining feature engineering and Spark ML components to generate Top-N job recommendations for users with limited interaction history in standalone mode.

Technologies

Big Data & Data Engineering: PySpark, Spark, Databricks, Azure Data Factory, ETL/ELT, Medallion/Lakehouse architecture, data modeling (Star/Snowflake), data-quality & schema-drift validation, Parquet, Pandas

Databases & Cloud: Advanced SQL · Postgres/Supabase/pgvector, Amazon RDS/Redshift, DynamoDB, ChromaDB · AWS (EC2/S3/RDS/Redshift) · Azure (ADF/Databricks/ADLS), Docker, CI/CD (Jenkins/Git), Vercel

Languages: Python (Advanced), SQL (Advanced), TypeScript, Java

AI/Other: LLM agentic workflows (LangGraph), RAG, hybrid retrieval (RRF), OCR, Python REST APIs (FastAPI/Flask), React/Next.js

Education

Masters in Data Science | University of Texas at Arlington Aug 2022 – May 2024

- Relevant Coursework: Machine Learning, Data Science, Data Analytics, Database Systems, Data Mining, Neural Networks, Probability and Statistics.

BTech in ECE | Andhra University Visakhapatnam, India Jun 2016 – Sep 2020